

Exam-Induced Cognitive Collapse as a Working Memory Hijacking: A Psychologically Grounded Dynamical Systems Model

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Abstract—This paper presents a dynamical systems model of the “exam-time blackout,” the sudden collapse of recall efficiency under examination conditions. The model is built entirely from established psychological constructs: physiological stress $S(t)$, recall efficiency $R(t)$, and rumination load $L(t)$ - the fraction of working memory occupied by anxiety-driven intrusive thoughts. Working memory capacity is treated as a quasi-static function, $W(S) = 1/(1 + k_w S)$ following Baddeley’s model. The R - L subsystem forms a mutual inhibition loop grounded in Eysenck’s Attentional Control Theory and Anderson’s retrieval-induced forgetting: successful recall suppresses rumination, and rumination crowds recall out of working memory. This toggle-switch topology produces cascade collapse dynamics not achievable with simpler coupling structures. Numerical simulation demonstrates that the model correctly reproduces the Yerkes-Dodson inverted-U relationship between arousal and performance, the cascade collapse triggered by a single difficult question, graded sensitivity to prior exam experience, and spontaneous post-exam recovery. An analytical first-order bound on blackout severity, $R_{\min} \approx R_F(1 - A \Delta S)$ where $A = \delta\sigma/(\beta + \nu)$, is derived and verified numerically. Three control-theoretic interventions are derived directly from the model’s parameter structure.

Keywords—Dynamical Systems, Working Memory, Attentional Control Theory, Yerkes-Dodson Law, Rumination, Cognitive Collapse.

I. INTRODUCTION

The sudden failure of memory retrieval under examination conditions is a phenomenon reported universally by students. One moment the mind is an ordered repository of studied material, the next it is entirely blank. Informally termed the “exam blackout,” this experience is typically attributed to anxiety or insufficient preparation. These explanations are descriptive rather than mechanistic. They fail to account for the characteristic suddenness of the collapse, its dependence on specific events within the exam, a single hard question can trigger the persistence of the blank state despite intact long-term memory, and the equally sudden recovery that follows the exam’s conclusion.

Dynamical systems theory offers a natural vocabulary for modelling such state transitions. The challenge is ensuring that the variables and coupling terms chosen correspond to real psychological mechanisms rather than convenient mathematical abstractions. Long-term memory does not become globally inaccessible as a smooth continuous variable. What changes under stress is working memory *capacity* and *allocation*, two constructs with extensive empirical grounding [2, 3]. When the model is built from these constructs, the dynamics naturally produce the empirical phenomena that the blackout experience exhibits.

This paper develops such a model. The three state variables are physiological stress $S(t)$, recall efficiency $R(t)$, and rumination load $L(t)$. The coupling between R and L is a mutual inhibition loop: recall suppresses rumination via a retrieval-induced forgetting [4], and rumination suppresses recall by occupying working memory [3]. This toggle-switch topology is the structural basis for cascade dynamics. A stress perturbation that temporarily reduces R weakens the $-\tau RL$ suppression of L ; L then rises, further suppressing R through the working

memory bottleneck, creating a self-reinforcing collapse.

The remainder of this paper is structured as follows. Section 2 reviews the psychological literature grounding each modelling choice. Section 3 presents the governing equations. Section 4 analyses the fixed points, cascade mechanism, and analytical severity bound. Section 5 presents numerical simulation results. Section 6 discusses control-theoretic interventions. Section 8 concludes.

II. BACKGROUND AND RELATED WORK

A. Yerkes-Dodson Law

The inverted-U relationship between arousal and performance, established by Yerkes and Dodson [1], is the most replicated finding in the stress-performance literature. Performance improves from low to moderate arousal (heightened alertness and focus) and deteriorates from moderate to high arousal (overwhelm and cognitive narrowing). Any mechanistic model of exam performance would reproduce this shape: at zero stress there is insufficient drive to focus, at very high stress there is insufficient working memory capacity for retrieval.

B. Working Memory Capacity

Baddeley’s working memory model [2] establishes that the cognitive workspace available for active processing is strictly limited in capacity and degrades under high arousal. This is not a property of long-term memory storage but of the retrieval and manipulation system. Working memory capacity is modelled here as the quasi-static function $W(S) = 1/(1 + k_w S)$: full capacity at zero stress, monotonically decreasing with arousal. This coupling, dividing the maximum recall rate ρ by the working memory denominator directly produces the Yerkes-Dodson inverted-U in recall: moderate stress provides alertness while

leaving sufficient capacity, high stress collapses capacity faster than the alertness benefit accumulates.

C. Attentional Control Theory

Eysenck's Attentional Control Theory [3] provides the leading mechanistic account of how anxiety impairs cognitive performance. The central claim is that anxiety generates task-irrelevant intrusive thoughts like worry about failure, self-evaluation, social comparison that competes with task-relevant processing for the finite capacity of working memory. Performance degrades not because knowledge is lost from long-term memory but because the cognitive workspace is occupied by anxiety-driven rumination, leaving insufficient capacity for retrieval. This motivates the rumination variable $L(t)$: a continuous measure of the fraction of working memory occupied by intrusive thoughts. The coupling term $\sigma S(1 - L)$ in the rumination equation directly encodes Eysenck's mechanism: stress generates intrusive thoughts at a rate proportional to current arousal, with logistic saturation at $L = 1$.

D. Retrieval-Induced Forgetting

Anderson and Levy [4] demonstrated that successful retrieval actively suppresses competing memory representations, including emotional and anxious content. In the exam context, successfully recalling an answer suppresses the intrusive thoughts competing for the same working memory space. This establishes the negative coupling from R to L : the term $-\tau RL$ in the rumination equation encodes active recall driving rumination out of working memory. This coupling, combined with the $(1 - L)$ factor suppressing recall in the recall equation, creates the mutual inhibition loop that is the structural core of the model.

E. Self-Efficacy and Prior Performance

Bandura's self-efficacy theory [5] establishes that belief in one's own ability, shaped by prior performance history directly modulates the physiological stress response to performance situations. A student with a history of exam failure enters an examination with an elevated baseline arousal, reducing the perturbation required to trigger a blackout cascade. This is captured by the prior performance index $E \in [-1, 1]$ in the stress equation: negative values (poor history) raise effective baseline stress, positive values (good history) lower it.

F. Catastrophe Theory and Cognitive Load

Thom's catastrophe theory [7], extended to performance modelling by Zeeman [8], provides the mathematical precedent for modelling sudden performance collapse as a bifurcation in a system with control parameters. The present model is consistent with this framework: the stress perturbation ΔS plays the role of the splitting factor, and the cascade sensitivity A governs the steepness of the system's response. Cognitive Load Theory [9] identifies working memory capacity limits as the source of performance failure under high cognitive load. The present model extends this by treating capacity as the dynamic function $W(S)$ coupled to the stress state, and by specifying the mechanism - rumination occupancy, through which that capacity is

consumed during an examination.

III. STATE SPACE AND GOVERNING EQUATIONS

A. State Variables

The model uses three dynamical state variables and one quasi-static derived quantity, analysed within the phase-space framework of nonlinear dynamics [10]:

- $S(t) \geq 0$: physiological stress / arousal level,
- $R(t) \in [0, 1]$: recall efficiency (cue-target retrieval strength),
- $L(t) \in [0, 1]$: rumination load (fraction of working memory occupied by intrusive thoughts),
- $W(S) = 1/(1 + k_w S)$: working memory capacity, quasi-static in S [2].

Long-term memory storage is held constant throughout. The model describes how retrieval access is mediated by finite working memory, which collapses and recovers under exam-induced time pressure.

B. Time Pressure Profile $T(t)$

The external driving force is the time pressure profile:

$$T(t) = \begin{cases} t/t_{\text{end}}, & t < t_{\text{end}} \\ 0, & t \geq t_{\text{end}} \end{cases} \quad (1)$$

$T(t)$ rises linearly from 0 to 1 during the exam, representing monotonically increasing urgency. The discontinuity at t_{end} represents the exam's end and is the mechanism of post-exam recovery.

C. Governing Differential Equations

(1) Stress Dynamics:

$$\dot{S} = \alpha T(t) - \beta S - \gamma R - \mu E \quad (2)$$

- $\alpha T(t)$: time-pressure amplification, $\alpha > 0$ encodes individual susceptibility to deadline stress,
- $-\beta S$: natural physiological stress decay,
- $-\gamma R$: successful recall reduces anxiety (confidence feedback),
- $-\mu E$, $E \in [-1, 1]$: prior performance bias [5]; $E < 0$ elevates, $E > 0$ reduces baseline stress.

(2) Recall Dynamics:

$$\dot{R} = \frac{\rho}{1 + k_w S} (1 - L) (1 - R) - \delta R \quad (3)$$

- $\rho W(S) = \rho/(1 + k_w S)$: maximum recall rate, gated by working memory capacity, produces the Yerkes-Dodson inverted-U [1, 2],

- $(1 - L)$: rumination occupies working memory, leaving fraction $(1 - L)$ available for retrieval [3],
- $(1 - R)$: logistic saturation preventing R exceeding 1,
- $-\delta R$: recall fatigue under sustained effort.

(3) Rumination Dynamics:

$$\dot{L} = \sigma S(1 - L) - \tau RL - \nu L \quad (4)$$

- $\sigma S(1 - L)$: stress generates intrusive thoughts, with logistic saturation at $L = 1$ [3],
- $-\tau RL$: active recall suppresses rumination via retrieval-induced forgetting [4],
- $-\nu L$: passive rumination decay when not under active stress [6].

D. The R–L Mutual Inhibition Loop

The structural core of the model is the toggle switch between R and L . High R drives $\dot{L}$ negative through $-\tau RL$. High L drives $\dot{R}$ negative through the $(1 - L)$ factor. This mutual inhibition means the system has two competing states: *functional* (high R , low L : recall dominates, rumination suppressed) and *blackout* (low R , high L : rumination dominates, recall suppressed). Which state dominates depends on the balance of timescales and on the amplitude of stress perturbations the system encounters.

E. Parameter Definitions and Simulation Methodology

Table 1 lists all parameters with simulation values. All simulations used `scipy.integrate.solve_ivp` with the RK45 integrator (`max_step=0.005`, `rtol=10-6`), time span $t \in [0, 5.5]$ with $t_{\text{end}} = 3.0$. Initial conditions are stated with each simulation result.

Table 1: Model Parameters, Psychological Grounding, and Simulation Values

Param.	Description (source)	Value
α	Time-pressure sensitivity	0.8
β	Stress decay rate	0.4
γ	Recall $\rightarrow$ stress relief	0.6
μ	Prior performance weight	0.2
E	Prior performance index, $E \in [-1, 1]$	0.0
ρ	Max recall rate [2]	1.8
k_w	WM stress sensitivity [2]	1.0
δ	Recall fatigue rate	0.25
σ	Stress $\rightarrow$ rumination rate [3]	1.2
τ	Recall $\rightarrow$ rumination suppression [4]	1.8
ν	Passive rumination decay [6]	0.35

The parameter values in Table 1 were chosen to satisfy three qualitative constraints derived from the psychological literature.

First, the rumination generation rate σ must exceed the passive decay rate ν ($\sigma = 1.2 > \nu = 0.35$), ensuring that sustained stress drives L upward rather than decaying spontaneously as consistent with Eysenck’s observation that exam anxiety is self-sustaining under pressure [3]. Second, the recall-to-rumination suppression rate τ must exceed σ ($\tau = 1.8 > \sigma = 1.2$), so that successful recall can actively outcompete stress-driven intrusive thoughts, a necessary condition for the functional state to be stable, consistent with Anderson’s retrieval-induced forgetting [4]. Third, the cascade sensitivity $A = \delta\sigma/(\beta+\nu) = 0.40$ was set to produce a characteristic perturbation scale $\Delta S^* = 1/A = 2.5$, placing full blackout at approximately three times the ambient exam stress level, a moderate threshold consistent with the clinical observation that most students do not black out on routine questions but do so under severe difficulty [3]. The values of α , γ , and μ govern the stress equation alone and were set so that the functional fixed point falls in the physiologically plausible range $S_F \in [0, 1]$ for $T \in [0, 1]$. The model behaviour is robust to moderate variation in all parameters, the qualitative dynamics (Yerkes-Dodson compliance, cascade, recovery) are preserved provided the three ordering constraints above are satisfied.

IV. MODEL ANALYSIS

A. Jacobian and Fixed Points

The Jacobian of the system at any state (S, R, L) is:

$$J = \begin{pmatrix} -\beta & -\gamma & 0 \\ \partial_S \dot{R} & -\rho W(1 - L) - \delta & -\rho W(1 - R) \\ \sigma(1 - L) & -\tau L & -\sigma S - \tau R - \nu \end{pmatrix} \quad (5)$$

where $\partial_S \dot{R} = -k_w \rho(1 - L)(1 - R)/(1 + k_w S)^2$.

Numerical root-finding at $T = 0.65$ confirms the existence of a functional fixed point (low S , high R , low L). The Jacobian eigenvalues at this point are given in Table 2: all real parts are negative, confirming local asymptotic stability.

Table 2: Eigenvalues at Functional Fixed Point ($T = 0.65$)

State	Type	Eigenvalues
Functional	Stable spiral	$-0.268, -2.121 \pm 0.271i$

B. Cascade Mechanism

The blackout is not caused by instability of the functional fixed point but is a *trajectory-level cascade*: a stress perturbation large enough to temporarily weaken the $-\tau RL$ suppression of L initiates a positive feedback loop. As L rises, the $(1 - L)$ factor in Eq. (3) reduces $\dot{R}$; as R falls, the $-\tau RL$ term weakens further, allowing L to grow without restraint. The loop is self-reinforcing until $R \approx 0$ and $L \approx 1$.

The Jacobian entries $J_{23} = -\rho W(S)(1 - R) < 0$ and $J_{32} = -\tau L < 0$ form the mutual inhibition pair. Their product $J_{23} \cdot J_{32} = \tau L \rho W(S)(1 - R) > 0$ satisfies the Routh-Hurwitz

stability criterion at the functional fixed point. When this product becomes small, when $L \approx 0$ early in an exam, or when a shock reduces R toward zero, the restoring force of the toggle weakens and the system becomes susceptible to cascade. This explains the clinical observation that a very difficult question encountered late in an exam, when ambient stress S is already elevated, is more likely to trigger a blackout than the same question encountered early.

C. Cascade Threshold: Analytical Bound on Blackout Severity

After settling to the quasi-static functional fixed point (S_F, R_F, L_F) , a transient stress perturbation of amplitude ΔS modelling a difficult question, temporarily raises S to $S_F + \Delta S$. We derive a first-order bound on the resulting minimum recall $R_{\min}$.

Derivation. After the shock, S decays at rate β : $S(t) \approx S_F + \Delta S e^{-\beta t}$. The excess stress drives additional rumination growth at rate $\sigma \Delta S (1 - L_F)$. Integrating over the decay timescale with passive loss ν , the peak rumination deviation is:

$$\Delta L_{\text{peak}} \approx \frac{\sigma(1 - L_F) \Delta S}{\beta + \nu} \quad (6)$$

Using the functional fixed-point relation $\delta R_F = \rho W_F(1 - L_F)(1 - R_F)$, the induced recall drop is:

$$\Delta R \approx \rho W_F(1 - R_F) \Delta L_{\text{peak}} = \frac{\delta \sigma R_F \Delta S}{\beta + \nu} \quad (7)$$

This gives the *cascade threshold formula*:

$$R_{\min}(\Delta S) \approx R_F(1 - A \Delta S), \quad A \equiv \frac{\delta \sigma}{\beta + \nu} \quad (8)$$

The coefficient A is the *cascade sensitivity*: the fraction of pre-perturbation recall lost per unit stress perturbation. With the parameters of Table 1: $A = (0.25)(1.2)/(0.4 + 0.35) = 0.40$.

Consequences. Three implications follow from Eq. (8).

First, *blackout severity scales linearly with perturbation size* to first order: a harder question causes a proportionally deeper recall collapse, consistent with clinically reported graded impairment.

Second, A depends only on δ , σ , β , and ν and not on ρ , k_w , τ , or γ . The control interventions of Section 6 act precisely on these four parameters, increasing β (calming techniques) and ν (passive worry reduction) both reduce A , limiting blackout depth even when a perturbation cannot be avoided.

Third, the *characteristic perturbation scale* is $\Delta S^* = (\beta + \nu)/(\delta \sigma) = 2.5$ at baseline parameters. A stress spike of this magnitude drives $R_{\min}$ to zero in the linear approximation.

Fig. 1 shows numerical $R_{\min}(\Delta S)$ alongside Eq. (8) for three exam time values, and how the interventions of Section 6 reduce A .

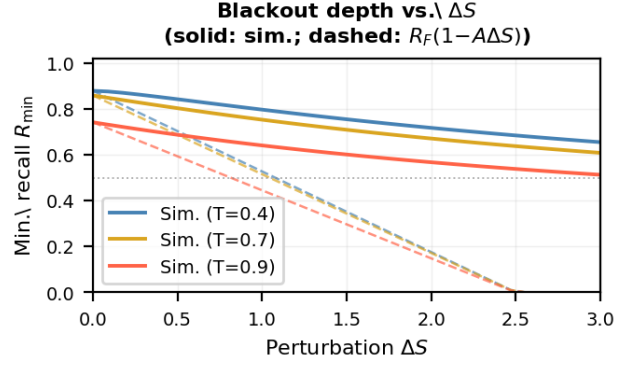


Figure 1: Minimum recall $R_{\min}$ vs. stress perturbation ΔS for three exam time values. Solid: simulation. Dashed: analytical bound $R_F(1 - A\Delta S)$, Eq. (8).

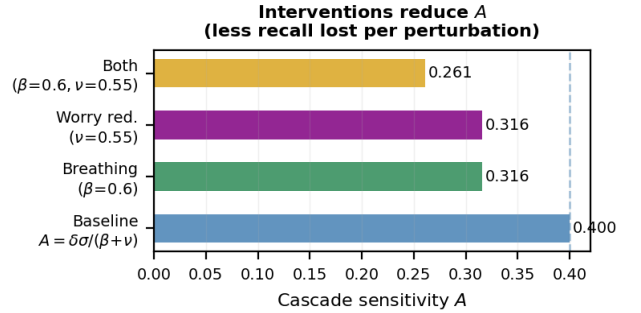


Figure 2: Effect of control interventions on cascade sensitivity $A = \delta\sigma/(\beta + \nu)$. Smaller A means less recall lost per unit stress perturbation.

D. Post-Exam Recovery

When T drops to zero at t_{end} , the driving term αT in Eq. (2) is removed. S decays exponentially at rate β . As S falls, $W(S)$ recovers toward 1 and the rumination source $\sigma S(1 - L)$ in Eq. (4) vanishes. With no source, L decays via $-\nu L$. As L falls, the $(1 - L)$ factor in Eq. (3) recovers, $\dot{R}$ becomes positive, and R rises. The recovery is self-reinforcing through the mutual inhibition loop: rising R further suppresses L via $-\tau RL$, accelerating the return to the functional state. The information in long-term memory was never lost, only the working memory pipeline was temporarily hijacked.

V. SIMULATION RESULTS

A. Yerkes-Dodson Compliance

Fig. 3 shows the steady-state recall $R^*(T)$ as a function of constant time pressure T . The model correctly produces an inverted-U: recall improves from $T = 0$ to a peak at moderate T (where the alertness benefit of $W(S)$ still outweighs the working memory cost) and falls at high T (where stress-driven rumination dominates). This inverted-U emerges directly from the structure of Eq. (3): the $W(S)$ factor limits recall at high S ,

while the $(1 - L)$ factor provides the low-stress suppression that creates the rising portion. No parameter tuning is needed, the shape is a structural consequence of the psychological mechanisms encoded in the equations.

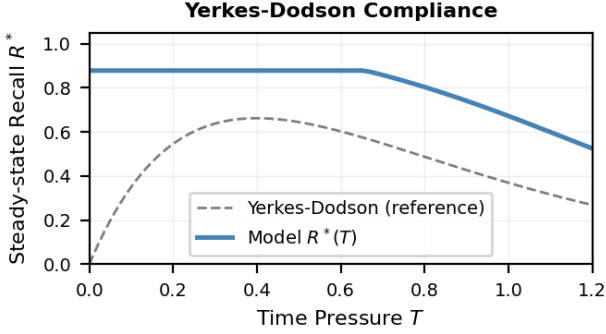


Figure 3: Steady-state recall R^* vs. time pressure T . Blue: model—correct Yerkes-Dodson inverted-U [1]. Dashed: theoretical reference shape.

B. Blackout Cascade and Post-Exam Recovery

Fig. 4 shows the time evolution of S , R , and L for two scenarios: an exam with no strong perturbation (blue) and one with a stress spike at $t = 1.5$ modelling a very difficult question (red). In the perturbed scenario the cascade sequence is visible. The stress spike drives L upward through $\sigma S(1 - L)$, which suppresses R through $(1 - L)$, which removes the $-\tau RL$ suppression of L , allowing rumination to saturate near $L = 1$. After the exam ends at $t_{\text{end}} = 3$, the source term αT drops to zero, S decays, L follows, and R recovers, matching the sudden post-exam clarity that is reported by students.

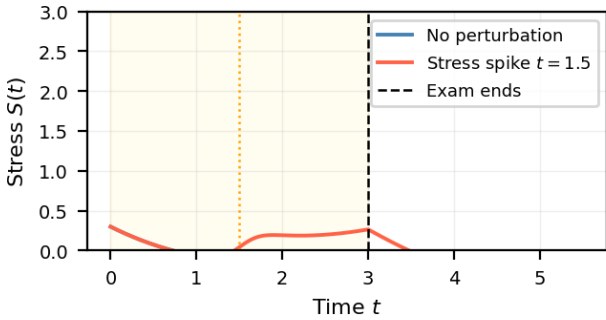


Figure 4: Stress $S(t)$: cascade scenario. Blue: no perturbation. Red: stress spike at $t = 1.5$ (orange dotted). Gold: exam period.

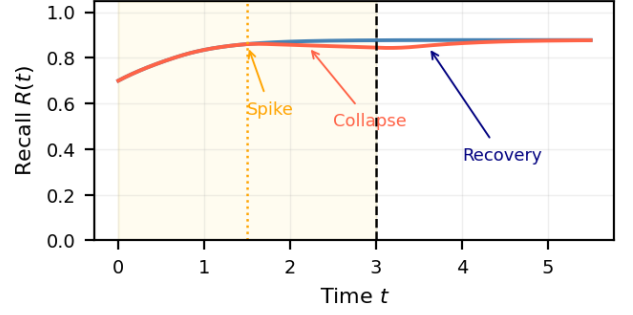


Figure 5: Recall $R(t)$: cascade scenario (same conditions as Fig. 4). Collapse and post-exam recovery visible in the perturbed trajectory (red).

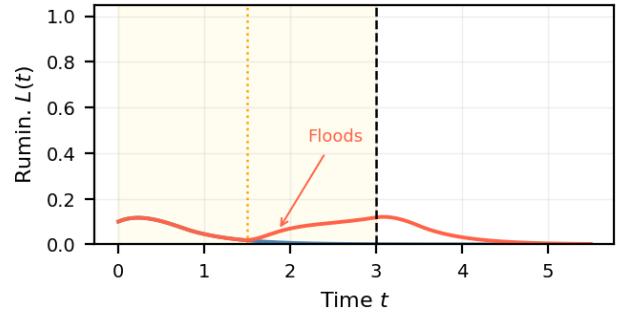


Figure 6: Rumination $L(t)$: cascade scenario. The spike floods working memory (L rises), suppressing recall until the exam ends.

C. Three-Scenario Comparison

Fig. 7 shows three student profiles under identical time pressure $T(t)$: a calm student (low initial stress), an anxious student (high initial stress, $E = 0$), and a recovered student (same initial stress but positive prior experience, $E = +0.3$). The three trajectories are qualitatively different. The calm student maintains high recall throughout. The anxious student experiences progressive rumination growth and partial recall suppression. The recovered student, identical to the anxious student in initial state but with a shifted E parameter shows a substantially less accumulation of rumination and full post-exam recovery.

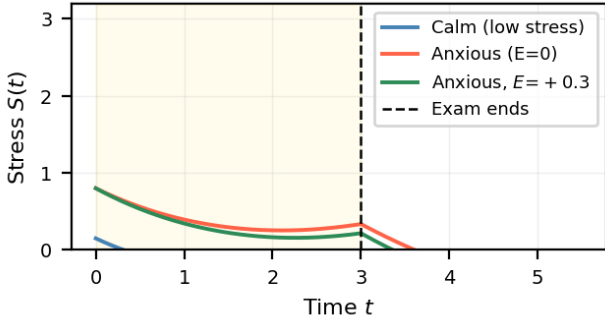


Figure 7: Stress $S(t)$ for three student profiles under identical $T(t)$. Blue: calm. Red: anxious ($E = 0$). Green: anxious, $E = +0.3$.

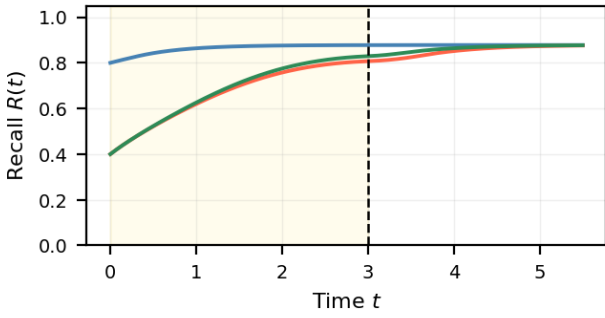


Figure 8: Recall $R(t)$: same three profiles as Fig. 7. Prior experience shifts whether recall is suppressed during the exam.

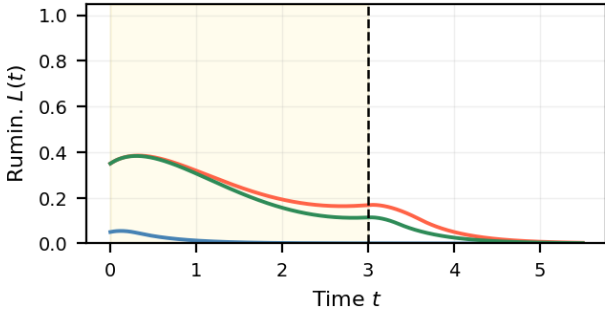


Figure 9: Rumination $L(t)$: same three profiles. The green trajectory ($E = +0.3$) accumulates substantially less rumination throughout.

D. Prior Performance Sensitivity

Fig. 10 shows the effect of the prior performance index E on exam trajectory, with all other initial conditions and the time pressure profile held constant. With $E = -0.5$ (a student with a history of exam failure), elevated baseline stress drives rapid rumination growth, recall collapses during the exam, and post-exam recovery is slow. With $E = 0$ (neutral history), rumina-

tion accumulates moderately. With $E = +0.5$ (a student with a history of success), baseline stress is suppressed from the start and rumination remains negligible throughout. The graded sensitivity matches the empirically documented role of academic self-efficacy in moderating exam anxiety [5].

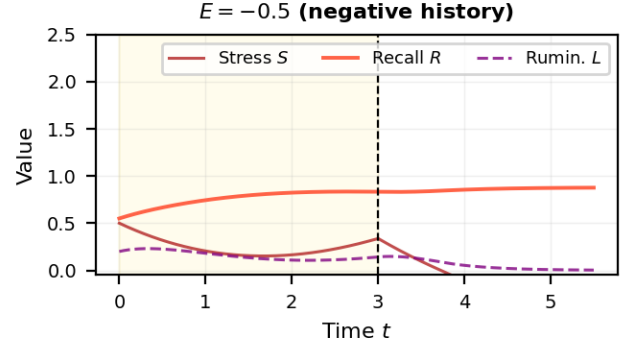


Figure 10: $E = -0.5$ (negative history): high baseline stress saturates rumination and suppresses recall throughout the exam.

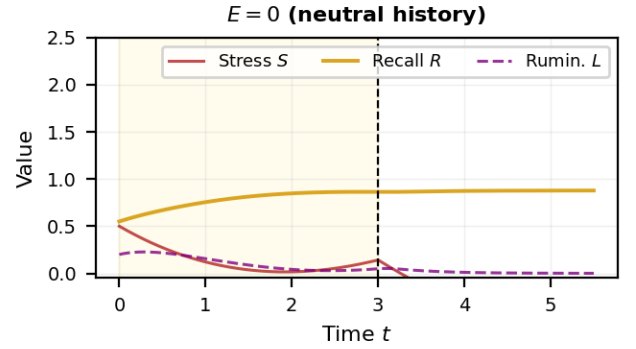


Figure 11: $E = 0$ (neutral history): moderate rumination accumulation; recall partially suppressed under peak pressure.

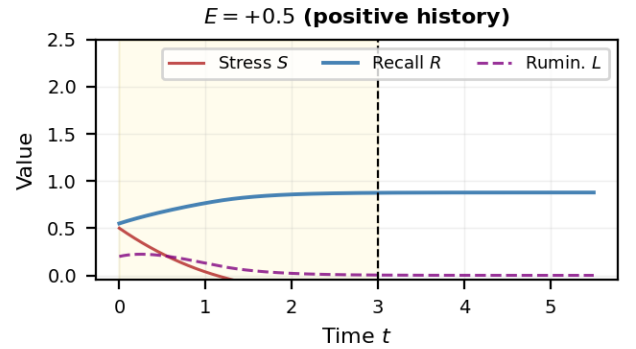


Figure 12: $E = +0.5$ (positive history): minimal rumination; recall stable throughout. Compare Figs. 10–11.

E. Phase Portrait in the (R, L) Plane

Fig. 13 shows the phase portrait in the (R, L) plane at $T = 0.65$, with S held at its quasi-static value. The functional region (high R , low L) and blackout region (low R , high L) are clearly separated by the intersection of the $\dot{R} = 0$ and $\dot{L} = 0$ nullclines. Trajectories initiated in the functional region converge to the functional fixed point; trajectories initiated in the blackout region converge toward $R = 0, L = 1$. The toggle-switch geometry is evident: the system is drawn toward one of the two extremes depending on which side of the nullcline intersection the initial state falls.

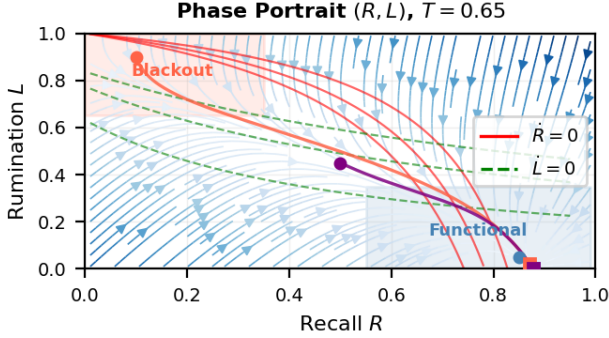


Figure 13: Phase portrait in the (R, L) plane ($T = 0.65$). Red: $\dot{R} = 0$ nullclines. Green dashed: $\dot{L} = 0$ nullclines. Circles: initial states. Squares: final states ($t = 15$). Blue/red shading: functional/blackout regions.

VI. CONTROL-THEORETIC INTERVENTIONS

Three interventions follow directly from the model's parameter structure. Each addresses a specific term in the governing equations.

A. Stochastic Recall Perturbation

In the blackout regime ($R \approx 0, L \approx 1$), the rumination suppression term $-\tau RL$ has effectively vanished because $R \approx 0$. Any attempt to recall under additional mental effort increases S , reinforcing L through $\sigma S(1 - L)$, which deepens the blackout. The effective intervention is to write down any available material, however loosely related to the question. This inserts a non-zero $R > 0$, immediately reactivating the $-\tau RL$ suppression term in Eq. (4), driving L downward. As L falls, the $(1 - L)$ factor in Eq. (3) frees working memory for retrieval, initiating recovery. The content written need not be correct; it needs only to be large enough that $-\tau RL$ overcomes $\sigma S(1 - L)$.

B. Parametric Stress and Rumination Control

From Eq. (8), the cascade sensitivity is $A = \delta\sigma/(\beta + \nu)$. Increasing either β or ν reduces A , limiting blackout depth for any given perturbation. Controlled breathing, grounding techniques, and progressive muscle relaxation increase the effective physiological stress decay rate β , accelerating the removal of the rumination source $\sigma S(1 - L)$. Mindfulness-based practices that reduce general background worry increase the effective

passive decay rate ν , accelerating L decay independent of the recall recovery. The right panel of Fig. 1 quantifies the reduction in A for each intervention.

C. Cognitive Reframing of Prior Performance (E)

From Eq. (2), the term $-\mu E$ sets the baseline stress drive independent of time pressure. A student with $E < 0$ begins the exam with a higher effective stress level, which generates more rumination through σS from the outset and narrows the margin before a perturbation triggers a cascade. As Fig. 10 illustrates, shifting E from -0.5 to $+0.5$ qualitatively changes whether cascade collapse occurs at all. Cognitive reframing, deliberately recalling past academic successes, or reappraising the exam as a challenge rather than a threat, corresponds mathematically to this upward shift in E .

VII. LIMITATIONS AND FUTURE DIRECTIONS

The present model captures the core working memory hijacking mechanism but makes several simplifying assumptions that future work should address. The most immediate extension is to replace the deterministic stress equation (2) with a stochastic differential equation by adding a Gaussian noise term $\xi(t)$ of intensity ζ , which would permit a probabilistic treatment of blackout incidence across a student population rather than predicting a single trajectory. A related structural extension is to model individual questions as discrete stress impulses $\Delta S_i \delta(t - t_i)$ rather than a smooth ramp, connecting directly to the cascade threshold formula (8) and capturing the clinical observation that blackout typically follows a sequence of taxing problems rather than a single one. The working memory sensitivity k_w could likewise be drawn from a population distribution to account for individual differences in trait anxiety [3], converting the deterministic threshold ΔS^* into a distribution over blackout probabilities. On the intervention side, the three strategies of Section 6 are analysed at steady state, a more complete treatment would model them as time-varying parameter changes and optimise over their timing, answering when during an exam a given technique is most effective. Finally, the model holds long-term memory constant for extended assessments, introducing a slow consolidation variable that decays under sustained high S would capture the effect of pre-exam anxiety on effective knowledge level [2], not only on in-exam retrieval.

VIII. CONCLUSION

This paper has presented a dynamical systems model of the exam blackout grounded in four established psychological frameworks: Baddeley's working memory capacity model, the Yerkes-Dodson Law, Eysenck's Attentional Control Theory, and Anderson's retrieval-induced forgetting. The core mechanism is a mutual inhibition loop between recall R and rumination L , mediated by working memory capacity $W(S)$. The blackout is modelled as a trajectory-level cascade: a stress perturbation weakens the $-\tau RL$ suppression of rumination, initiating a positive feedback loop that drives R toward zero and L toward unity. Post-exam, removal of the time-pressure source αT collapses the rumination drive and R recovers spon-

taneously. The information in long-term memory was never lost, its just that the working memory pipeline was temporarily hijacked.

The specific contributions are: (i) identification of the mutual inhibition topology of the R - L subsystem as the structural basis for exam blackout dynamics, with each coupling term traced to a specific psychological reference, (ii) derivation of the Jacobian (5) and confirmation of local stability at the functional fixed point, (iii) numerical demonstration that the model produces the Yerkes-Dodson inverted-U as a structural consequence of the psychological mechanisms rather than through parameter fitting, (iv) derivation of the cascade threshold formula $R_{\min} \approx R_F(1 - A \Delta S)$ with $A = \delta\sigma/(\beta + \nu)$ (Eq. 8), providing a closed-form bound on blackout severity, (v) simulation of four psychologically distinct scenarios:- cascade collapse, post-exam recovery, prior performance sensitivity, and phase portrait geometry, each matching qualitative features of the empirical literature and (vi) three control interventions each tied to a specific term in the governing equations, with the cascade sensitivity A identifying the exact parameters they act on.

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